

1 Conservation and dissipation of energy

1.1 Changes in energy stores

Learning objectives

After this topic, you should know:

- the ways in which energy can be stored
- how energy can be transferred
- the changes in energy stores that happen when an object falls
- the energy transfers that happen when a falling object hits the ground without bouncing back.

On the move

Cars, buses, planes, and ships all use fuels as chemical energy stores. They carry their own fuel. Electric trains use energy transferred from fuel in power stations. Electricity transfers energy from the power station to the train.



Figure 1 The French Train à Grande Vitesse electric train can reach speeds of more than 500 km/hour

Energy can be stored in different ways and is transferred by heating, waves, an electric current, or when a force moves an object. Here are some examples:

- Chemical energy stores include fuels, foods, or the chemicals found in batteries. The energy is transferred during chemical reactions.
- Kinetic energy stores describe the energy an object has because it is moving.
- Gravitational potential energy stores are used to describe the energy stored in an object because of its position, such as an object above the ground.
- Elastic potential energy stores describe the energy stored in a springy object when you stretch or squash it.
- Thermal energy stores describe the energy a substance has because of its temperature.

Energy can be transferred from one store to another. In a torch, the torch's battery pushes a current through the bulb. This makes the torch bulb emit light, and also get hot (Figure 2).

When an electric kettle is used to boil water, the current in the kettle's heating element transfers energy to the thermal energy store of the water and the kettle.

When an object is thrown into the air, the object slows down as it goes up. Here, energy is transferred from the object's kinetic energy store to its gravitational potential energy store.

You can show the energy transfers by using a flow diagram:

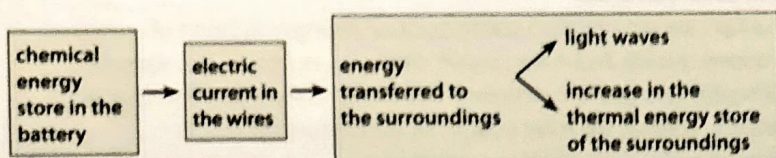


Figure 2 Changes in energy stores in a torch lamp

Energy transfers

When an object starts to fall freely, it speeds up as it falls. The force of gravity acting on the object causes energy to be transferred from its gravitational potential energy store to its kinetic energy store.

Look at Figure 3. It shows an object that hits the floor with a thud. All of the energy in its kinetic energy store is transferred by heating to the thermal energy store of the object and the floor, and by sound waves moving away from the point of impact. The amount of energy transferred by sound waves is much smaller than the amount of energy transferred by heating.

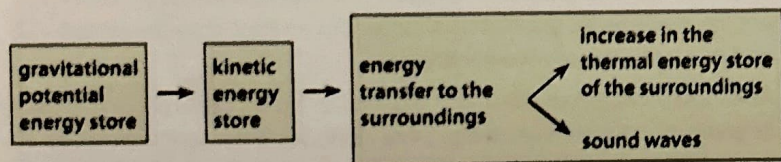


Figure 4 An energy transfer diagram for an object when it falls and when it hits the ground

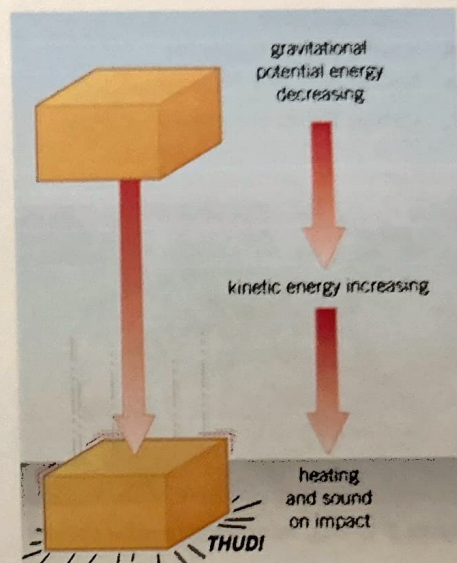


Figure 3 An energetic drop. On impact, energy is transferred to the thermal energy store of the surroundings by heating and by sound waves

- Describe the changes to energy stores that take place when:
 - a ball falls in air [2 marks]
 - an electric heater is switched on. [2 marks]
- List two different objects you could use to light a room if you have a power cut. For each object, describe the energy transfers and changes to energy stores that occur when it lights up the room. [4 marks]
 - Which of the two objects in a is:
 - easier to obtain energy from? [1 mark]
 - easier to use? [1 mark]
- Describe the changes in energy stores of an electric train as it:
 - moves up a hill at constant speed [2 marks]
 - approaches a station and brakes to a halt. [2 marks]
- Describe the changes in energy stores that take place when food is heated in a microwave oven. [2 marks]

Key points

- Energy can be stored in a variety of different energy stores.
- Energy is transferred by heating, by waves, by an electric current, or by a force when it moves an object.
- When an object falls and gains speed, its store of gravitational potential energy decreases and its kinetic energy store increases.
- When a falling object hits the ground without bouncing back, its kinetic energy store decreases. Some or all of its energy is transferred to the surroundings – the thermal energy store of the surroundings increases, and energy is also transferred by sound waves.

P1.2 Conservation of energy

Learning objectives

After this topic, you should know:

- what conservation of energy is
- why conservation of energy is a very important idea
- what a closed system is
- how to describe the changes to energy stores in a closed system.



Figure 1 Energy transfers on a roller coaster

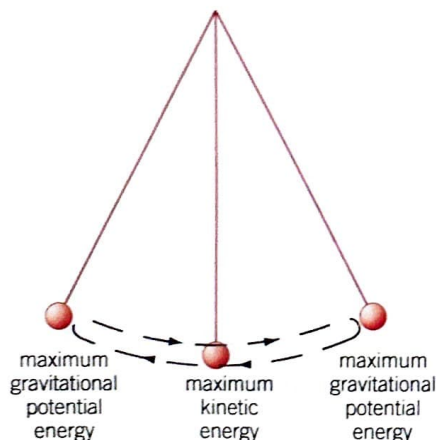


Figure 2 A pendulum in motion. As the pendulum swings down and towards the centre, its gravitational potential energy store decreases as its kinetic energy store increases. As the pendulum moves upwards and away from the centre, its gravitational potential energy store increases as its kinetic energy store decreases

At the funfair

Funfairs are very exciting places because changes to stores of energy happen quickly. As a roller coaster climbs an incline, its gravitational potential energy store increases. This energy is then transferred to other energy stores as the roller coaster races downwards.

As the roller coaster descends, its gravitational potential energy store decreases. Most of this energy is transferred to its kinetic energy store, which therefore increases. However, some energy is transferred to the thermal energy store of the surroundings by air resistance and friction, and some energy is transferred by sound waves.

Investigating pendulums

When changes to energy stores happen, does the total amount of energy stay the same? You can investigate this question with a simple pendulum.

Figure 2 shows a pendulum bob swinging from side to side.

As it moves towards the middle, energy is transferred by the force of gravity from its gravitational potential energy store to its kinetic energy store. So its gravitational potential energy store decreases and its kinetic energy store increases.

As it moves away from the middle, its kinetic energy store decreases and its gravitational potential energy store increases. If the air resistance on the bob is very small the bob will reach the same height on each side.

- Describe the changes to energy stores that take place in the bob when it goes from one side at maximum height to the other side at maximum height.
- Explain why it is difficult to mark the exact height the pendulum bob rises to. Suggest how you could make your judgement of height more accurate.

Conservation of energy

The pendulum in Figure 2 would probably keep on swinging for ever if it was in a vacuum because there would be no air resistance acting on it, and so no energy would be transferred from any of its energy stores. There would be no net change to the energy stored in the system. Because of this, it would be an example of a closed system.

A system is an object or a group of objects. Scientists have done lots of tests and have concluded that the total energy of a closed system is always the same before and after energy transfers to other energy stores within the closed system.

This important result is known as the principle of conservation of energy. It says that:

energy cannot be created or destroyed.

Energy can be stored in various ways. For example:

- when a rubber band is stretched, its elastic potential energy store increases
- when an object is lifted, its gravitational potential energy store is increased.

Bungee jumping

What energy transfers happen to a bungee jumper after jumping off the platform?

- When the rope is slack, energy is transferred from the gravitational potential energy store to the kinetic energy store as the jumper accelerates towards the ground due to the force of gravity.
- When the rope tightens, it slows the bungee jumper's fall. This is because the force of the rope reduces the speed of the jumper. The jumper's kinetic energy store decreases and the rope's elastic potential energy store increases as the rope stretches. Eventually the jumper comes to a stop – the energy that was originally in the kinetic energy store of the jumper has all been transferred into the elastic potential energy store of the rope.

After reaching the bottom, the rope recoils and pulls the jumper back up. As the jumper rises, the energy in the elastic potential energy store of the rope decreases and the bungee jumper's kinetic energy store increases until the rope becomes slack. After the rope becomes slack, and at the top of the ascent, the bungee jumper's kinetic energy store decreases to zero. The bungee jumper's gravitational potential energy store increases throughout the ascent.

The bungee jumper doesn't return to the original height. This is because some energy was transferred to the thermal energy store of the surroundings by heating as the rope stretched and then shortened again.

- 1 When a roller coaster gets to the bottom of a descent, describe the energy transfers and changes to energy stores that happen if:
 - a the brakes are applied to stop it [4 marks]
 - b it goes up and over a second hill. [5 marks]
- 2 a A ball dropped onto a trampoline returns to almost the same height after it bounces. Describe the energy transfers and changes to the energy stores of the ball from the point of release to the top of its bounce. [5 marks]
 - b Describe the energy stores of the ball at the point of release compared with its energy stores at the top of its bounce. [1 mark]
 - c Describe how you would use the test in a to see which of three trampolines is the bounciest. [5 marks]
- 3 One exciting fairground ride acts like a giant catapult. The capsule in which you are strapped is fired high into the sky by the rubber bands of the catapult. Explain the changes to the energy stores that take place in the ride as you move upwards. [4 marks]

Study tip

Never use the terms 'movement energy' or 'motion energy' – only 'kinetic energy'.

Bungee jumping

You can try out the ideas about energy transfers during a bungee jump using the experiment shown in Figure 3.

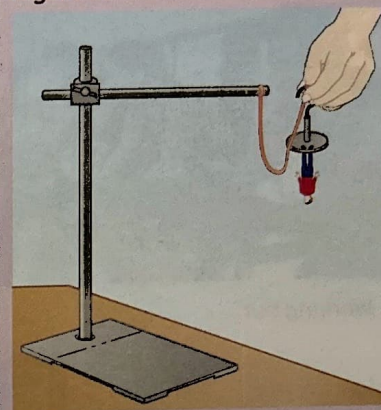


Figure 3 Testing a bungee jump

Safety: Make sure the stand is secure. Protect feet and bench from falling objects.

Key points

- Energy cannot be created or destroyed.
- Conservation of energy applies to all energy changes.
- A closed system is an isolated system in which no energy transfers take place out of or into the energy stores of the system.
- Energy can be transferred between energy stores within a closed system. The total energy of the system is always the same, before and after, any such transfers.

P1.3 Energy and work

Learning objectives

After this topic, you should know:

- what work means in science
- how work and energy are related
- how to calculate the work done by a force
- what happens to work done to overcome friction.



Figure 1 Working out

Worked example

A builder pushed a wheelbarrow a distance of 5.0 m across flat ground with a force of 50 N. How much work was done by the builder?

Solution

$$\begin{aligned}\text{work done} &= \text{force applied} \times \text{distance moved} \\ &= 50 \text{ N} \times 5.0 \text{ m} \\ &= 250 \text{ J}\end{aligned}$$



Figure 2 Pulling a lorry

Working out

In a fitness centre or a gym, you have to work hard to keep fit. Lifting weights and pedalling on an exercise bike are just two ways to keep fit. Whichever way you choose to keep fit, you have to apply a force to move something. So the work you do causes a transfer of energy.

When an object is moved by a force, **work** is done on the object by the force. So the force transfers energy to the object. The amount of energy transferred to the object is equal to the work done on it. For example, to raise an object, you need to apply a force to it to overcome the force of gravity on it. If the work you do on the object is 20 J, the energy transferred to it must be 20 J. So its gravitational potential energy store increases by 20 J.

$$\text{energy transferred} = \text{work done}$$

The work done by a force depends on the size of the force and the distance moved. One joule of work is done when a force of one newton causes an object to move a distance of one metre in the direction of the force. To calculate the work done by a force when it causes displacement of an object, use this equation:

$$\begin{array}{lll}\text{work done, } W &= \text{force applied, } F \times \text{distance moved along the line} \\ (\text{joules, J}) & (\text{newtons, N}) & \text{of action of the force, } s \\ & & (\text{metres, m})\end{array}$$

Superhuman force

Imagine pulling a lorry over 40 m. On level ground, a pull force of about 2000 N is needed. The work done by the pulling force is 80 kJ (= 2000 N × 40 m). Very few people can manage to pull with such force. Don't even try it, though. The people who can do it are very, very strong and have trained specially for it.

Doing work

Carry out a series of experiments to calculate the work done in performing the tasks below. Use a newton-meter to measure the force applied, and use a metre ruler to measure the distance moved.

- 1 Drag a small box a measured distance across a rough surface.
- 2 Repeat the test above with two rubber bands wrapped around the box (Figure 3).

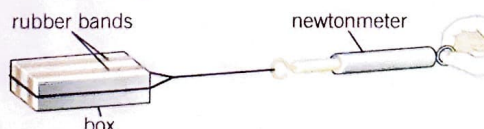


Figure 3 At work

- What is the resolution of your measuring instruments? Repeat your tests and comment on the precision of your repeat measurements. Can you be confident about the accuracy of your results?



Friction at work

Work done to overcome friction is mainly transferred to thermal energy stores by heating.

- 1 If you rub your hands together vigorously, they become warm. Your muscles do work to overcome the friction between your hands. The work you do is transferred as energy that warms your hands.
- 2 Brake pads on a vehicle become hot if the brakes are applied for too long. Friction between the brake pads and the wheel discs opposes the motion of the wheel. The force of friction does work on the brake pads and the wheel discs. As a result, energy is transferred from the kinetic energy store of the vehicle to the thermal energy store of the brake pads and the wheel discs. This makes them become hot and transfer energy by heating to the thermal energy store of the surrounding air.
- 3 Meteorites are small objects from space that enter the Earth's atmosphere and fall to the ground. As they pass through the atmosphere, friction caused by air resistance acts upon them. This results in energy being transferred from the meteorite's gravitational potential energy and kinetic energy stores to the meteorite's thermal energy store, causing the meteorite to heat up. If a meteorite becomes hot enough, it glows and becomes visible as a 'shooting star'. Very small objects can burn up completely. The surface of a space vehicle is designed to withstand the very high temperatures caused by this friction when it re-enters the Earth's atmosphere.

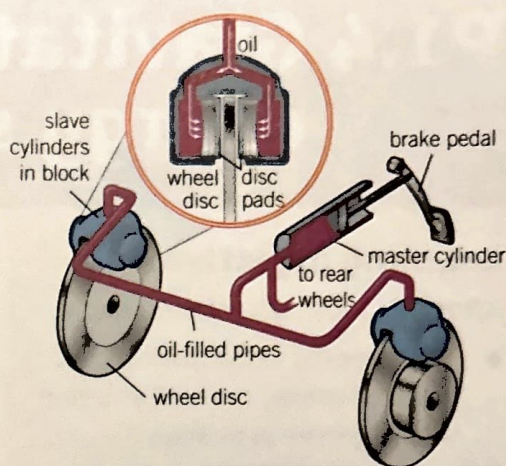


Figure 4 Disc brakes

- 1 a State what happens to the energy transferred:
 - i by a rower rowing a boat [1 mark]
 - ii by an electric motor used to raise a car park barrier. [1 mark]
 b Calculate how much work is done when a force of 3000 N pulls a truck through a distance of 50 m in the direction of the force. [1 mark]
- 2 A car is brought to a standstill when the driver applies the brakes.
 - a Explain why the brake pads become warm. [2 marks]
 - b The car travelled a distance of 20 metres after the brakes were applied. The braking force on the car during this time was 7000 N. Calculate the work done by the braking force. [1 mark]
- 3 a Calculate the work done when:
 - i a force of 20 N makes an object move 4.8 m in the direction of the force [1 mark]
 - ii an object of weight 80 N is raised through a height of 1.2 m. [1 mark]
 b When a cyclist brakes, his kinetic energy store is reduced from 1400 J to zero in a distance of 7.0 m. Calculate the braking force. [2 marks]
- 4 A student pushes a box at a steady speed a distance of 12 m across a level floor.
 - a The student applied a horizontal force of 25 N to the box. Calculate the work done by the student. [1 mark]
 - b Describe the energy transfers and changes to energy stores as the box moves. [3 marks]

Study tip

If you calculate the work done, this is equal to the energy transferred.

Key points

- Work is done on an object when a force makes the object move.
- Energy transferred = work done.
- Work done is $W = F s$ where F is the force and s is the distance moved (along the line of action of the force).
- Work done to overcome friction is transferred as energy to the thermal energy stores of the objects that rub together and to the surroundings.

P1.4 Gravitational potential energy stores

Learning objectives

After this topic, you should know:

- what happens to the gravitational potential energy stores of an object when it moves up or down
- why an object moving up has an increase in its gravitational potential energy store
- why it is easier to lift an object on the Moon than on the Earth
- how to calculate the change in gravitational potential energy of an object when it moves up or down.

Worked example

A student of weight 300 N climbs on a platform that is 1.2 m higher than the floor. Calculate the increase in her gravitational potential energy store.

Solution

$$\text{Increase of } E_p = 300 \text{ N} \times 1.2 \text{ m} = 360 \text{ J}$$

Worked example

A 2.0 kg object is raised through a height of 0.4 m. Calculate the increase in the gravitational potential energy store of the object. The gravitational field strength of the Earth at its surface is 9.8 N/kg.

Solution

$$\begin{aligned} \text{Gain of } E_p &= \text{mass} \times \text{gravitational field strength} \times \text{height gain} \\ &= 2.0 \text{ kg} \times 9.8 \text{ N/kg} \times 0.4 \text{ m} \\ &= 7.8 \text{ J} \end{aligned}$$

Changes in gravitational potential energy stores

Every time you lift an object up, you do some work. Some of your muscles transfer energy from the chemical energy store in the muscle to the gravitational energy store of the object. In calculations we refer to the energy in this store as gravitational potential energy E_p .

The force you need to lift an object at constant velocity is equal and opposite to the gravitational force on the object. So the upward force you need to apply to it is equal to the object's weight. For example, you need a force of 80 N to lift a box of weight 80 N.

- When an object is moved upwards, the energy in its gravitational potential energy store increases. This increase is equal to the work done on it by the lifting force to overcome the gravitational force on the object.
- When an object moves down, the energy in its gravitational potential energy store decreases. This decrease is equal to the work done by the gravitational force acting on it.

The work done when an object moves up or down depends on:

- 1 how far it is moved vertically (its change of height)
- 2 its weight.

Using work done = force applied $\times$ distance moved in the direction of the force:

$$\begin{array}{l} \text{change in object's gravitational} \\ \text{potential energy store} \\ \text{(joules, J)} \end{array} = \begin{array}{l} \text{weight} \\ \text{(newtons, N)} \end{array} \times \begin{array}{l} \text{change of height} \\ \text{(metres, m)} \end{array}$$

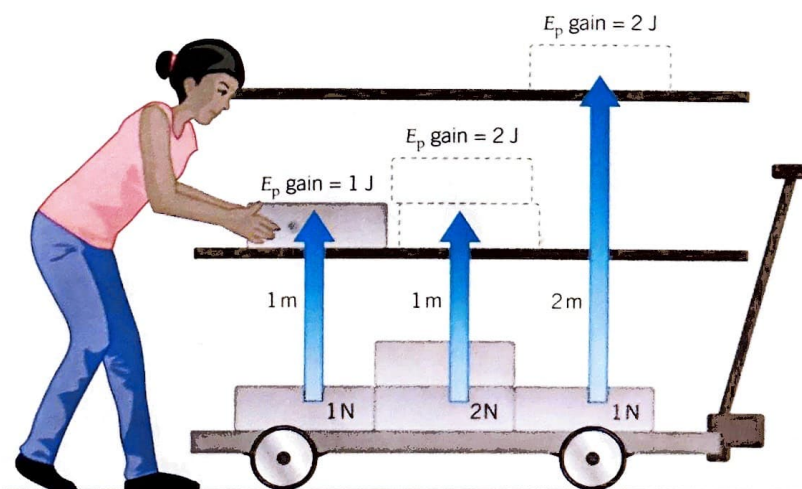


Figure 1 Using joules

Gravitational potential energy stores and mass

Astronauts on the Moon can lift objects much more easily than they can on the Earth. This is because the gravitational field strength on the Moon's surface is only about a sixth of the gravitational field strength on the Earth's surface.

You have previously learnt that the weight of an object in newtons is equal to its mass $\times$ the gravitational field strength. So, when an object is lifted or lowered, the change to its gravitational potential energy store is equal to its weight $\times$ its change of height. Therefore:

$$\begin{array}{l} \text{change of} \\ \text{gravitational} \\ \text{potential} \\ \text{energy store, } \Delta E_p \\ \text{(joules, J)} \end{array} = \begin{array}{l} \text{mass, } m \\ \text{(kilograms, kg)} \end{array} \times \begin{array}{l} \text{gravitational} \\ \text{field strength, } g \\ \text{(newtons per} \\ \text{kilogram, N/kg)} \end{array} \times \begin{array}{l} \text{change of} \\ \text{height, } \Delta h \\ \text{(metres, m)} \end{array}$$

Study tip

Watch out for objects going up a slope. To calculate the increase in their gravitational potential energy stores, you need the vertical height gained, *not* the distance along the slope.

Study tip

In physics, the greek letter delta (Δ) is used to represent the phrase 'change in'. For example, Δh can be used in place of 'change in height'.

Stepping up

Measure your mass in kilograms using floor scales.

Step on and off a sturdy box or low platform.

Measure the height of the box.

Use the equation $\Delta E_p = m g \Delta h$, where $g = 9.8 \text{ N/kg}$, to calculate the energy transferred to your gravitational potential energy store when you stepped on the box.

Safety: Make sure the box is secure and that you feel comfortable to do this.



Key points

- The gravitational potential energy store of an object increases when it moves up and decreases when it moves down.
- The gravitational potential energy store of an object increases when it is lifted up because work is done on it to overcome the gravitational force.
- The gravitational field strength at the surface of the Moon is less than on the Earth.
- The change in the gravitational potential energy store of an object is $\Delta E_p = m g \Delta h$

- a** Describe the changes to the energy stores of a ball when it falls and rebounds without regaining its initial height. [4 marks]

b When a ball of weight 1.4 N is dropped from rest from a height of 2.5 m above a flat surface, it rebounds to a height of 1.7 m above the surface.

 - Calculate the total energy lost from the ball's energy stores when it reaches this maximum rebound height. [2 marks]
 - Name two causes of the energy transfer. [2 marks]
- A student of weight 450 N steps on a box of height 0.2 m.

a Calculate the increase in her gravitational potential energy store. [1 mark]

b Calculate the work done by the student if she steps on and off the box 50 times. [1 mark]
- a** A weightlifter raises a steel bar of mass 25 kg through a height of 1.8 m. Calculate the change to the gravitational potential energy store of the bar. The gravitational field strength at the surface of the Earth is 9.8 N/kg. [2 marks]

b The weightlifter then lowers the bar by 0.3 m and drops it so it falls to the ground. Assume that air resistance is unimportant. Calculate the change to its gravitational potential energy store in this fall. [2 marks]
- You use energy when you hold an object stationary in your outstretched hand. Suggest what happens to the energy that must be supplied to your muscles to keep them contracted. [3 marks]

P1.5 Kinetic energy and elastic energy stores

Learning objectives

After this topic, you should know:

- what the amount of energy in a kinetic energy store depends on
- how to calculate the amount of energy in a kinetic energy store
- what an elastic potential energy store is
- how to calculate the amount of energy in an elastic potential energy store.

The energy an object has because of its motion depends on its mass and speed. This energy is called kinetic energy.

Investigating kinetic energy stores

Figure 1 shows how you can investigate how the kinetic energy store of an object depends on its speed.

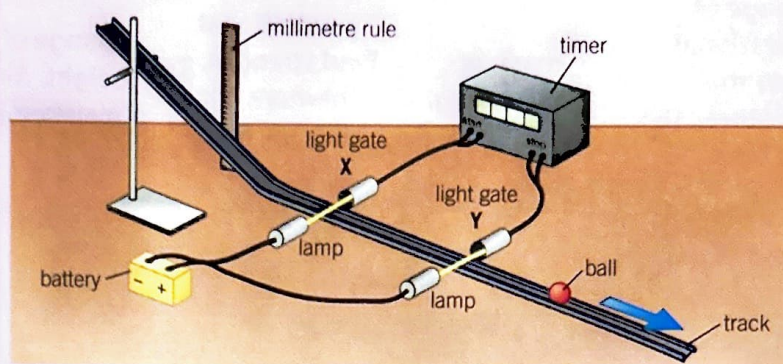


Figure 1 Investigating changes in kinetic energy stores

- 1 The ball is released on a slope from a measured height above the foot of the slope. You can calculate the decrease in its gravitational potential energy store by using the following equation:
$$\text{change in gravitational potential energy store} = \text{mass} \times \text{gravitational field strength} \times \text{change in height}.$$

Due to conservation of energy, this decrease in the gravitational potential energy store is matched by an equal increase in its kinetic energy store.
 - 2 The ball is timed, using light gates, over a measured distance between X and Y after the slope.
- Explain why light gates improve the quality of the data you can collect in this investigation.

Investigating a catapult

Use rubber bands to catapult a trolley along a horizontal runway. Find out how the speed of the trolley depends on how much the catapult is pulled back before the trolley is released. For example, see if the distance needs to be doubled to double the speed. Figure 1 shows how the speed of the trolley can be measured.

Safety: Take care to ensure you do this safely. Protect your hands and feet, and the bench, from falling trolleys.

Table 1 shows some sample results.

Table 1 Sample measurements for a ball of mass 0.5 kg

Height drop to foot of slope in m	0.05	0.10	0.16	0.20
Initial kinetic energy of ball in J	0.25	0.50	0.80	1.00
Time to travel 1.0 m from X to Y in s	0.98	0.72	0.57	0.50
Average speed of ball between X and Y in m/s	1.02			2.00

Work out the speed of the ball between X and Y in each case. The first and last values have been worked out for you. Can you see a link between speed and height drop? The results show that the greater the height drop, the faster is the speed. So it can be said that the kinetic energy store of the ball increases if the speed increases.

The kinetic energy equation

Table 1 shows that when the height drop is increased by four times from 0.05 m to 0.20 m, the speed doubles. The height drop is directly proportional to the speed squared, or (speed)². Because the height drop is a measure of the ball's kinetic energy store, it can be said that the ball's kinetic energy store is directly proportional to the square of its speed.

The amount of energy in the kinetic energy store of an object can be calculated using the kinetic energy equation below:

$$\text{kinetic energy, } E_k = \frac{1}{2} \times \text{mass, } m \times \text{speed}^2, v^2$$

(joules, J) (kilograms, kg) (metres per second, m/s)²

Using elastic potential energy

When you stretch a rubber band or a spring, the work you do is stored in it as **elastic potential energy**.

Figure 2 shows how the force F needed to stretch a spring varies with its extension e . The graph obeys the equation for Hooke's Law $F = ke$, where k is the spring constant.

For a spring stretched to an extension e , we can calculate the energy in its elastic potential energy store using the equation below:

$$\text{elastic potential energy, } E_e = \frac{1}{2} \times \text{spring constant, } k \times \text{extension}^2, e^2$$

(joules, J) (newtons per metre, N/m) (metres, m)²

Worked example

Calculate the kinetic energy stored in a vehicle of mass 500 kg moving at a speed of 12 m/s.

Solution

$$\begin{aligned} \text{Kinetic energy} &= \frac{1}{2} m v^2 \\ &= 0.5 \times 500 \text{ kg} \times (12 \text{ m/s})^2 \\ &= \mathbf{36\,000 \text{ J}} \end{aligned}$$

Go further!

In Figure 2, the force F increases as the extension e is increased. The average force when the spring is extended to extension e is $\frac{1}{2} F$, where $F = ke$. Therefore, the energy stored in the spring = work done = average force extension = $\frac{1}{2} F e = \frac{1}{2} k e^2$.

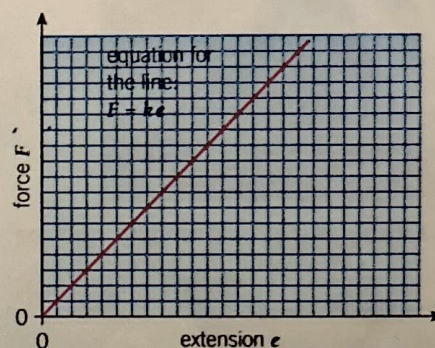


Figure 2 Force versus extension for a spring. The spring constant k is the force per unit extension of the spring

Key points

- The energy in the kinetic energy store of a moving object depends on its mass and its speed.
- The kinetic energy store of an object is $E_k = \frac{1}{2} m v^2$
- Elastic potential energy is the energy stored in an elastic object when work is done on the object.
- The elastic potential energy stored in a stretched spring is $E_e = \frac{1}{2} k e^2$, where e is the extension of the spring.

- a** Calculate the kinetic energy store of:

 - a vehicle of mass 500 kg moving at a speed of 12 m/s [2 marks]
 - a football of mass 0.44 kg moving at a speed of 20 m/s. [2 marks]

b Calculate the velocity of a 500 kg vehicle with twice the kinetic energy store as calculated in **a i**. [3 marks]
- a** A catapult is used to fire an object into the air. Describe the energy transfers when the catapult is:

 - stretched [2 marks]
 - released. [2 marks]

b An object of weight 2.0 N fired vertically upwards from a catapult reaches a maximum height of 5.0 m. Calculate:

 - the increase in the gravitational potential energy store of the object [1 mark]
 - the speed of the object when it left the catapult. [4 marks]
- A car moving at a constant speed has 360 000 J in its kinetic energy store. When the driver applies the brakes, the car stops in a distance of 100 m.

 - Calculate the force that stops the vehicle. [3 marks]
 - The speed of the car was 30 m/s when its kinetic energy store was 360 000 J. Calculate its mass. [3 marks]
- A mobility aid to assist walking uses a steel spring to store energy when the walker's foot goes down, and it returns energy as the foot is lifted. The spring has a spring constant of 250 N/m. Calculate the elastic potential energy stored in the spring when its extension is 0.21 m. [2 marks]

P1.6 Energy dissipation

Learning objectives

After this topic, you should know:

- what is meant by useful energy
- what is meant by wasted energy
- what eventually happens to wasted energy
- whether energy is still as useful after it is used.



Figure 1 Using energy on a running machine

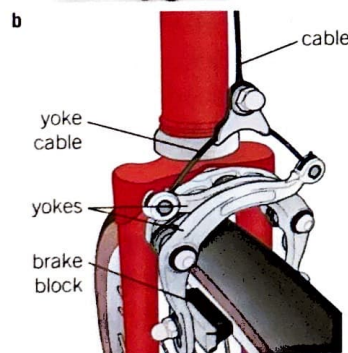


Figure 2 Friction in action: **a** Using a drill, **b** Braking on a bicycle

Energy for a purpose

Where would people be without machines? Washing machines are used to clean your clothes. Machines in factories are used to make the things you buy. You might use exercise machines in a gym to keep yourself fit, and machines are used to get you from place to place.

A machine transfers energy for a purpose. Friction between the moving parts of a machine causes the parts to warm up. So, not all of the energy supplied to a machine is usefully transferred. Some of the energy is wasted.

- **Useful energy** is energy transferred to where it is wanted in the way that is wanted.
- **Wasted energy** is the energy that is not usefully transferred.

Whenever energy is transferred for a purpose in any system, some of the energy is transferred usefully. The rest is **dissipated** (spreads out) and may be stored in less useful ways. This energy is described as wasted energy because it is not transferred as useful energy. For example, when a jet plane takes off, its engines transfer energy from the chemical energy store in the fuel. Some of this energy increases the kinetic energy and the gravitational potential energy stores of the plane, which is the useful energy transfer pathway. The rest is wasted energy because some of it heats the plane and the surroundings and some is transferred to the surroundings by sound waves created by the engine vibrations.

Investigating friction

Friction in machines always causes energy to be wasted. Figure 2 shows two examples of friction in action.

In Figure 2a, friction acts between the drill bit and the wood. The bit becomes hot as it bores into the wood. Energy is transferred by an electric current to the thermal energy stores, heating up the drill bit and the wood.

When a bike rider brakes, friction acts between the brake blocks and the wheels (Figure 2b). Energy is wasted as it is transferred from the kinetic energy stores of the bike and the cyclist, to the thermal energy stores of the brake blocks and the wheels, which are heated by friction.

Friction at work

Next time you use a running machine, think about what happens to the energy transferred by your muscles. As you are exercising, energy is transferred to the thermal energy store of your muscles (so you get hot) and to the thermal energy store of the machine by the force you exert to overcome the friction in the machine.

Spreading out

- Wasted energy is dissipated (spreads out) to the surroundings, for example, the gears of a car get hot because of friction when the car is running. Here, energy is transferred from the kinetic energy store of the gear box to the thermal energy stores of the gear box and the surrounding air. The thermal energy stores of the gear box and the surrounding air therefore increase, as do their respective temperatures.
- Useful energy is eventually transferred to the surroundings too. For example, the useful energy supplied to turn the wheels of a car is eventually transferred from the kinetic energy stores of the wheels to the thermal energy stores of the tyres by heating – increasing the thermal energy stores of the tyres. This energy is then transferred to the thermal energy store of the road and surrounding air.
- Energy becomes less useful the more it spreads out. For example, the hot water from a central heating boiler in a building is pumped through pipes and radiators. The thermal energy store of the hot water decreases as it transfers energy by heating to the thermal energy stores of the radiators – heating the rooms in the building. But the energy supplied to heat these rooms will eventually be transferred to the surrounding air.

1 Copy and complete the table below.

Energy transfer by	Useful energy output	Wasted energy output
a An electric fan heater	warms the air and surrounding objects	
b A television		
c An electric kettle		
d Headphones		

[4 marks]

2 Describe what would happen, in terms of energy transfer and changes to energy stores, to:

- a a gear box that was insulated so it could not transfer energy by heating to the surroundings [3 marks]
 - b the running shoes of a jogger if the shoes are well insulated [2 marks]
 - c a blunt electric drill bit if you use it to drill into hard wood [2 marks]
 - d the metal wheel discs of the brakes of a car when the brakes are applied. [2 marks]
- 3 a Describe the energy transfers and changes to energy stores of a pendulum as it swings from one side to the middle, and then to the opposite side. [4 marks]
- b Explain why a swinging pendulum eventually stops. [3 marks]
- 4 A freewheeling cyclist on a level road gradually stops moving. Describe the energy transfers and changes to the energy stores of the cyclist and the bicycle. [4 marks]

Key points

- Useful energy is energy in the place we want it and in the form we need it.
- Wasted energy is the energy that is not useful energy and is transferred by an undesired pathway.
- Wasted energy is eventually transferred to the surroundings, which become warmer.
- As energy dissipates (spreads out), it gets less and less useful.

P1.7 Energy and efficiency

Learning objectives

After this topic, you should know:

- what is meant by efficiency
- what is the maximum efficiency of any energy transfer
- how machines waste energy
- **H** how energy transfers can be made more efficient.

When you lift an object, energy stored in your muscles is transferred to the gravitational potential energy store of the object. The amount transferred depends on the object's weight and how high you lift it.

- Weight is measured in newtons (N). The weight of a 1 kilogram object on the Earth's surface is about 10 N.
- Energy is measured in joules (J). The energy needed to lift a weight of 1 newton by a height of 1 metre is equal to 1 joule.

The energy supplied to the device is called the input energy. The useful energy transferred by the device is called the useful output energy.

Because energy cannot be created or destroyed:

Input energy (energy supplied, J) = useful output energy (useful energy transferred, J) + energy wasted (J)

For any device that transfers energy:

$$\text{efficiency} = \frac{\text{useful output energy transferred by the device (J)}}{\text{total input energy supplied to the device (J)}}$$

Efficiency

Efficiency can be written as a decimal number (that is always less than 1) or as a percentage.

For example, a light bulb with an efficiency of 0.15 would radiate 15 J of energy as light for every 100 J of energy you supply to it.

- Its efficiency (as a number) = $\frac{15}{100} = 0.15$
- Its percentage efficiency = $0.15 \times 100\% = 15\%$



Worked example

An electric motor is used to raise an object. The object's gravitational potential energy store increases by 60 J when the motor is supplied with 200 J of energy by an electric current. Calculate the percentage efficiency of the motor.

Solution

Total energy supplied to the device = 200 J

Useful energy transferred by the device = 60 J

Percentage efficiency of the motor

$$\begin{aligned} &= \frac{\text{useful output energy transferred by the device}}{\text{total input energy supplied to the device}} \times 100\% \\ &= \frac{60 \text{ J}}{200 \text{ J}} \times 100\% = 0.30 \times 100\% = 30\% \end{aligned}$$

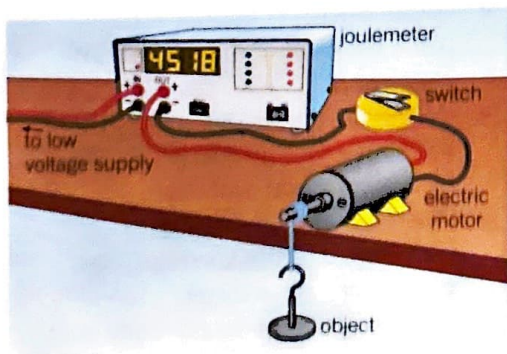


Figure 1 An electric winch

Efficiency limits

No device can be more than 100% efficient, because you can never get more energy from a machine than you put into it.

Investigating efficiency

Figure 1 shows how you can use an electric winch to raise a weight. You can use the joulemeter to measure the energy supplied.

- If you double the weight for the same increase in height, do you need to supply twice as much energy to do this task?

To calculate the change in the gravitational potential energy store of the weight use the following equation:

gravitational potential energy = weight in newtons $\times$ height increase in metres.



Use this equation and the joulemeter measurements to work out the percentage efficiency of the winch.

- Determine how the efficiency depends on the weight of the object being raised.

Plot a graph of your results, and use it to discuss how the efficiency of the winch changes with weight.

- Make some more measurements to find out whether lubricating the axle of the electric motor with a few drops of suitable oil makes the winch more efficient. Switch the motor off when you lubricate it.

Safety: Protect the floor and your feet. Stop the winch before the masses wrap round the pulley.

Study tip

The greater the percentage of energy that is usefully transferred in a device, the more efficient the device is.

Higher

Improving efficiency

Table 1 Increasing the efficiency of devices

	Why devices waste energy	How to reduce the problem
1	Friction between the moving parts causes heating.	Lubricate the moving parts to reduce friction.
2	The resistance of a wire causes the wire to get hot when a current passes through it.	In circuits, use wires with as little electrical resistance as possible.
3	Air resistance causes a force on a moving object that opposes its motion. Energy transferred from the object to the surroundings by this force is wasted.	Streamline the shapes of moving objects to reduce air resistance.
4	Sound created by machinery causes energy transfer to the surroundings.	Cut out noise (e.g., tighten loose parts to reduce vibration).

Study tip

Efficiency and percentage efficiency are numbers without units. The maximum efficiency is 1, or 100%, so if your calculation produces a number greater than this, it must be wrong.

- a** Compare the useful output energy by a machine and the input energy supplied to the machine. [1 mark]

b Explain why the percentage efficiency of a machine can never be:

i more than 100% [2 marks] **ii** equal to 100%. [4 marks]
- An electric motor is used to raise a weight. When you supply 60 J of energy to the motor, the weight gains 24 J of gravitational potential energy. Calculate:

a the energy wasted by the motor [1 mark]

b the efficiency of the motor. [2 marks]
- A machine is 25% efficient. If the total energy supplied to the machine is 3200 J, calculate how much useful energy can be transferred. [2 marks]
- An electric fan heater contains a motor that blows hot air into the room. Describe the changes in the energy stores of the heater. [4 marks]

Key points

- The efficiency of a device = $\frac{\text{useful energy transferred by the device}}{\text{total energy supplied to the device}} \times 100\%$.
- No energy transfer can be more than 100% efficient.
- Machines waste energy because of friction between their moving parts, air resistance, electrical resistance, and noise.
- **H** Machines can be made more efficient by reducing the energy they waste. For example, lubrication is used to reduce friction between moving parts.

P1.8 Electrical appliances

Learning objectives

After this topic, you should know:

- how energy is supplied to your home
- why electrical appliances are so useful
- what most everyday electrical appliances are used for
- how to choose an electrical appliance for a particular job.

Synoptic links

You should know that electricity is not an energy store, but rather it is a flow of charge that transfers energy from one energy store to another. You will learn more about electricity in Chapter P4.

Everyday electrical appliances

The energy you use in your home is mostly supplied by electricity, gas, and oil. Although all three of these energy supplies can be used for cooking and heating, your electricity supply is vital because you use electrical appliances for so many purposes every day. The charge that flows through these appliances transfers energy to them, which they then transfer usefully. But some of the energy you supply to them is wasted.



Figure 1 Electrical appliances – how many can you see in this photo?

Table 1 Comparing energy use in electrical appliances

Appliance	Useful energy	Energy wasted
Light bulb	Light emitted from the glowing filament.	Energy transfer from the filament heating the surroundings.
Electric heater	Energy heating the surroundings.	Light emitted from the glowing element.
Electric toaster	Energy heating bread.	Energy heating the toaster case and the air around it.
Electric kettle	Energy heating water.	Energy heating the kettle itself.
Hairdryer	Kinetic energy of the air driven by the fan.	Sound of fan motor (energy heating the motor heats the air going past it, so is not wasted).
	Energy heating air flowing past the heater filament.	Energy heating the hairdryer itself.
Electric motor	Kinetic energy of objects driven by the motor. Gravitational potential energy of objects lifted by the motor.	Energy heating the motor and energy transferred by the sound waves generated by the motor.

Clockwork radio

People without electricity supplies can now listen to radio programmes – thanks to the British inventor Trevor Baylis. In the early 1990s, he invented the clockwork radio. When you turn a handle on a clockwork radio, you wind up a clockwork spring in it, and increase the elastic potential energy of the spring. When the spring unwinds, energy from its elastic potential energy store is transferred to its kinetic energy store and it turns a small electric generator in the radio. It doesn't need batteries or mains electricity. So people in remote areas where there is no mains electricity can listen to their radios without having to walk miles for a replacement battery. But they do have to wind up the spring every time its store of energy has been used.



Figure 2 Clockwork radios are now mass produced and sold all over the world

Choosing an electrical appliance

You use electrical appliances for many purposes. Each appliance is designed for a specific purpose, and it should waste as little energy as possible. Suppose you were a rock musician at a concert. You would need appliances that transfer the variations in sound waves to electricity and then back to sound waves. But you wouldn't want the appliances to transfer lots of energy to the thermal energy store of the surroundings or themselves. See if you can spot some of these appliances in Figure 3.

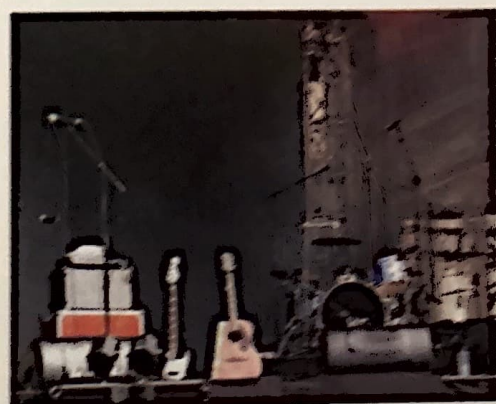


Figure 3 *On stage*

- 1 Match each electrical appliance in the list below with the energy transfer A or B it is designed to bring about.
- Energy transfer **A** Energy transferred by an electric current → energy transferred by sound waves
- B** Energy transferred by an electric current → kinetic energy store
- a** Electric drill [1 mark]
- b** Food mixer [1 mark]
- c** Electric bell [1 mark]
- 2 **a** Explain why a clockwork radio needs to be wound up before it can be used. [2 marks]
- b** Describe the changes in energy stores that take place in a clockwork radio when it is wound up and then switched on. [2 marks]
- c** Give one advantage and one disadvantage of a clockwork radio compared with a battery-operated radio. [2 marks]
- 3 An electric dishwasher heats water and sprays the hot water at the dishes. It then pumps the water out.
- a** Describe how energy is usefully used in the machine. [1 mark]
- b** Describe how energy is wasted by the machine. [2 marks]
- 4 A laptop battery stores energy with an efficiency of 80% when it is recharged from a low-voltage power supply. When it is connected to the laptop, it transfers energy to the laptop with an efficiency of 60%.
- a** Calculate the overall percentage efficiency of the laptop battery. [2 marks]
- b** Calculate the overall percentage of energy wasted. [1 mark]

Key points

- Electricity and gas and/or oil supply most of the energy you use in your home.
- Electrical appliances can transfer energy in the form of useful energy at the flick of a switch.
- Uses of everyday electrical appliances include heating, lighting, making objects move (using an electric motor), and producing sound and visual images.
- An electrical appliance is designed for a particular purpose and should waste as little energy as possible.

P1.9 Energy and power

Learning objectives

After this topic, you should know:

- what is meant by power
- how to calculate the power of an appliance
- how to calculate the efficiency of an appliance in terms of power
- how to calculate the power wasted by an appliance.

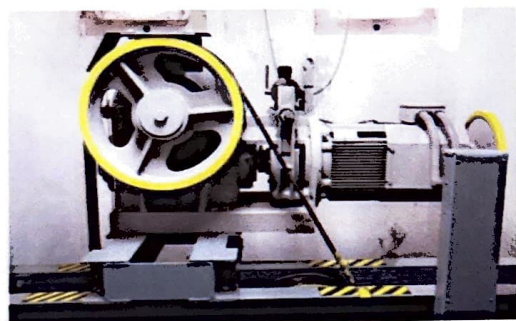


Figure 1 A lift motor

Worked example

A motor transfers 10 000 J of energy in 25 s. Work out its power.

Solution

$$P = \frac{E}{t}$$

$$P = \frac{10\,000\text{ J}}{25\text{ s}} = 400\text{ W}$$

Study tip

Always use the correct units in your calculations. Values that have been converted to micro (μ), milli (m), kilo (k), or mega (M) must be converted back to the SI unit. Standard form is useful when working with large numbers (See Maths skills 1b for more about standard form.).

When you use a lift to go up, a powerful electric motor pulls you and the lift upwards. The electric current through the lift motor transfers energy to the gravitational potential energy store of the lift when the lift goes up at a steady speed. You also get energy (from the electric current) transferred to the thermal energy store of the motor and the surroundings due to friction between the moving parts of the motor. In addition, energy is transferred to the thermal energy store of the surroundings by sound waves created by the lift machinery.

- The energy you supply to the motor per second is the **power** supplied to it.
- The more powerful the lift motor is, the faster it moves a particular load.

The more powerful an appliance is, the faster the rate at which it transfers energy.

The power of an appliance is measured in watts (W) or kilowatts (kW). 1 watt is equal to the rate of transferring 1 joule of energy in 1 second (i.e., $1\text{ W} = 1\text{ J/s}$)

1 kilowatt is equal to 1000 watts (i.e., 1000 joules per second or 1 kJ/s).

You can calculate power using the equation:

$$\text{power, } P \text{ (watts, W)} = \frac{\text{energy transferred to appliance, } E \text{ (joules, J)}}{\text{time take for energy to be transferred, } t \text{ (seconds, s)}}$$

Power ratings

Table 1 shows some typical values of power ratings for different energy transfers.

Table 1

Appliance	Power rating
A torch	1 W
An electric light bulb	100 W
An electric cooker	10 000 W = 10 kW (where 1 kW = 1000 watts)
A railway engine	1 000 000 W = 1 megawatt (MW) = 1 million watts
A Saturn V space rocket	100 MW
A very large power station	10 000 MW
World demand for power	10 000 000 MW
A star like the Sun	100 000 000 000 000 000 000 MW

Orders of magnitude

The power ratings in Table 1 are called 'order of magnitude' values. This means that the values are estimates to the nearest power of ten.

The symbol $\sim$ is used for an order of magnitude estimate. For example, if ten million people switch their electric kettles on at the same time (e.g., at half-time in a big TV football match), the jump in the demand for electric power would be $\sim 10\,000\text{ MW}$ (10^4 MW).

Muscle power

How powerful is a weightlifter?

A 30 kg dumbbell has a weight of 300 N. Raising it by 1 m would increase its gravitational potential energy store by 300 J. A weightlifter could lift it in about 0.5 seconds. The rate of energy transfer would be 600 J/s ($= 300 \text{ J} \div 0.5 \text{ s}$). So the weightlifter's power output would be about 600 W in total! The power output of two weightlifters could be compared by measuring how long they each take to lift the same weight through the same vertical height. The energy transferred is the same. So the one who takes the less time has a bigger power output.

Efficiency and power

For any appliance:

- its useful power out (or output power) is the useful energy per second transferred by it
- its total power in (or input power) is the energy per second supplied to it.

In Topic P1.7, you learnt that, for an appliance:

$$\text{efficiency} = \frac{\text{useful energy transferred by the device}}{\text{total energy supplied to it}} (\times 100\%)$$

Because power = energy per second transferred or supplied, this efficiency equation can be rewritten as

$$\text{efficiency} = \frac{\text{useful power out}}{\text{total power in}} (\times 100\%)$$

Wasted power

In any energy transfer, the energy wasted = the input energy supplied – the useful output energy. Because power is energy transferred per second:

$$\text{power wasted} = \text{total power in} - \text{useful power out}$$

- a** State which of the following is more powerful.

 - A torch bulb or a mains filament lamp. [1 mark]
 - A 3 kW electric kettle or a 10 000 W electric cooker. [1 mark]

b There are about 20 million occupied homes in England. If a 3 kW electric kettle was switched on in 1 in 10 homes at the same time, work out how much power would need to be supplied. [1 mark]
- The total power supplied to a lift motor is 5000 W. In a test, the motor transfers 12 000 J of energy in 20 s to the gravitational potential energy store of the lift.

 - Calculate how much energy is supplied to the motor by the current through it in 20 s. [2 marks]
 - Calculate its efficiency in the test. [2 marks]
- A machine has an input power rating of 100 kW. If the useful energy transferred by the machine in 50 seconds is 1500 kJ, calculate:

 - its output power in kilowatts [2 marks]
 - its percentage efficiency. [2 marks]
- a** Describe the energy transfers and changes to the energy stores of an electric hot-water shower when it is in operation. [5 marks]

b A 12 kW electric shower is used 4 times in a day for 20 minutes each time. Calculate the energy supplied to it by electricity in one day. [2 marks]



Figure 2 Muscle power

Worked example

The useful power out of an electric motor is 20 W, and the total power into it is 80 W. Calculate:

- the percentage efficiency of the motor
- the power wasted by the motor.

Solution

a percentage efficiency = $\frac{\text{useful power out}}{\text{total power in}} \times 100\%$

$$= \frac{20 \text{ W}}{80 \text{ W}} \times 100\% = 25\%$$

b wasted power = $80 \text{ W} - 20 \text{ W} = 60 \text{ W}$

Key points

- Power is rate of transfer of energy.
- The power of an appliance is $P = \frac{E}{t}$.
- efficiency of an appliance

$$= \frac{\text{useful power out}}{\text{total power in}} (\times 100\%)$$
- power wasted by an appliance = total power input – useful power output